DOI: https://doi.org/10.31640/LS-2025-1-05

UDC 616.61-008.6:577.1+616.1+616.8-005.4+616-092

L. D. Denova, *MD, PhD-student,*
ORCID ID: 0000-0002-5678-5885,
Shupyk National Healthcare University of Ukraine, Kyiv, Ukraine,
Department of Nephrology and Renal Replacement Therapy,
marbua18@gmail.com

S. V. Kushnirenko, *MD, Doctor of Medical Sciences, Professor,*
ORCID: 0000-0001-5518-7210,
Shupyk National Healthcare University of Ukraine, Kyiv, Ukraine,
Department of Nephrology and Renal Replacement Therapy,
stella-alex@i.ua

O. O. Rudenko, *MD, Neurologist,*
ORCID: 0009-0006-8794-5438,
Brovary Multiprofile Clinical Hospital, Brovary, Ukraine,
rudenkoolga443@gmail.com

SYSTEMIC ACTION OF UROMODULIN: EXTRA-RENAL EFFECTS AND CLINICAL SIGNIFICANCE

Abstract. *Uromodulin (UMOD) is the most abundant protein in human urine, traditionally regarded as a marker of tubular kidney function. However, modern scientific research increasingly indicates its multifaceted systemic action that extends far beyond the nephron. This protein is involved in complex mechanisms regulating vascular tone, uric acid metabolism, immune response, and endothelial function. Its association with the development of cardiovascular pathology, ischemic stroke, hyperuricemia, and metabolic syndrome has been established. This review article analyzes a wide spectrum of experimental and clinical sources, including meta-analyses, systematic reviews, genetic and metabolomic studies, which confirm the significant role of UMOD in the pathogenesis of extra-renal diseases. Uromodulin (UMOD) is the most abundant protein in human urine, traditionally regarded as a marker of the functional state of renal tubules. However, modern scientific studies increasingly indicate its multifaceted systemic action that extends far beyond the nephron. This protein is involved in complex mechanisms of vascular tone regulation, uric acid metabolism, immune response, and endothelial function. Its association with the development of cardiovascular pathology, ischemic stroke, hyperuricemia, and metabolic syndrome has been established. This review analyzes a wide range of experimental and clinical sources, including meta-analyses, systematic reviews, genetic and metabolomic studies, which confirm the significant role of UMOD in the pathogenesis of extra-renal diseases.* ***Conclusions:*** *The summarized data support considering UMOD as a promising biomarker and a potential tool in personalized medicine.*

Keywords: *uromodulin; uric acid; ischemic stroke; hypertension; extra-renal effects; biomarker.*

© Л. Д. Денова, С. В. Кушніренко, О. О. Руденко, 2025

INTRODUCTION

Uromodulin (UMOD) is one of the key proteins synthesized in the thick ascending limb of the loop of Henle. Its physiological function was long believed to be limited to ion transport and protection of the urinary tract from infections [1]. However, in recent years, our understanding of this molecule has significantly expanded. In vitro and in vivo studies, as well as large population-based cohorts, have revealed the systemic effects of UMOD, in particular on cardiovascular, neurological, and metabolic regulation [2, 3].

It is now established that UMOD is present not only in urine but also in blood, where it performs a number of biologically active functions. Its interaction with endothelial cells, the immune system, and metabolites such as uric acid opens new perspectives for understanding its role in the pathogenesis of systemic diseases. In particular, reduced UMOD levels are associated with an increased risk of arterial hypertension, ischemic stroke, cognitive impairment, and progression of chronic kidney disease [4–7].

Genetic studies further confirm the role of UMOD as a systemic marker: its variants are associated with the risk of developing hypertension, cerebrovascular *lesions, and metabolic syndrome [8, 9].*

Thus, uromodulin can be considered not only a renal marker but also a potential systemic biomarker and a target for directed intervention. This review systematizes current knowledge of the extra-renal effects of UMOD, focusing on its role in cardiovascular, cerebrovascular, and metabolic disorders. Special attention is paid to the evidence base of clinical, genetic, and metabolomic studies that demonstrate the multifaceted role of this protein in inter-organ interaction and the development of chronic non-communicable diseases.

The aim of this review article is to characterize the systemic, extra-renal effects of uromodulin, with a focus on its role in the pathogenesis of cardiovascular, cerebrovascular, and metabolic disorders, as well as to substantiate the potential of UMOD as a biomarker and therapeutic tool in personalized medicine.

UMOD and Cardiovascular Pathology

Data from large population studies indicate that low serum UMOD levels are associated with an increased risk of hypertension, cardiovascular events, and mortality. UMOD has been found to be associated with levels of vasoconstrictors such as CT-proET-1 and influences blood pressure regulation through HER2-dependent pathways [3, 4, 10]. This is supported by cohort studies involving patients with CKD, elderly individuals, and patients with diabetes [3, 7, 11–13] (*see Table 1*).

Table 1. UMOD: Renal and Extra-Renal Effects

№	Site of Action	Primary Action of UMOD	Clinical Significance
1	Kidneys	Ion transport, antimicrobial action	Biomarker of nephron function
2	Cardiovascular system	Vasoregulation, HER-2/EGFR signaling pathways	Reduction of cardiovascular risk
3	Central nervous system	Influence on endothelium, inflammation, hemodynamics	Decreased risk of ischemia and cognitive impairment
4	Metabolism	Impact on sUrAc excretion	Reduced risk of hyperuricemia, gout, and metabolic syndrome

Note. [1, 2, 5, 6–8, 10, 12, 15, 16, 42–48, 52].

Impact of Uromodulin on the Central Nervous System

Uromodulin shows involvement in mechanisms related to cerebrovascular regulation, particularly in experimental models of hypertension with a high risk of stroke. Its decreased excretion under conditions of excessive salt intake is accompanied by intracellular accumulation and alterations in cerebral hemodynamics [14–17, 26]. Proteomic changes in brain blood samples confirm the presence of UMOD during ischemic events.

Relationship Between UMOD and Uric Acid

Genetic variants of UMOD demonstrate an influence on uric acid transport and the development of hyperuricemia [19–21]. The absence or deficiency of UMOD activates compensatory mechanisms in the tubules, leading to increased uricemia and arterial pressure [23]. The biomarker significance of the UMOD/creatinine ratio has been confirmed in patients with gout and CKD [18, 24].

Uric Acid and Ischemic Stroke

The relationship between uric acid and ischemic stroke is actively studied in meta-analyses, systematic reviews, and cohort studies. Results indicate a dose-dependent association between UA levels and stroke severity or prognosis [27–44] (see Fig. 1). Additionally, in a study involving patients with ischemic stroke, elevated uric acid levels were associated with a higher risk of adverse outcomes during hospitalization [45]. Some authors highlight the "obesity paradox" related to UA [43], as well as the modifying effect of gender [42]. The role of UA has also been investigated in the context of post-stroke fatigue and cognitive impairment [46–48] (*see Fig. 2*).

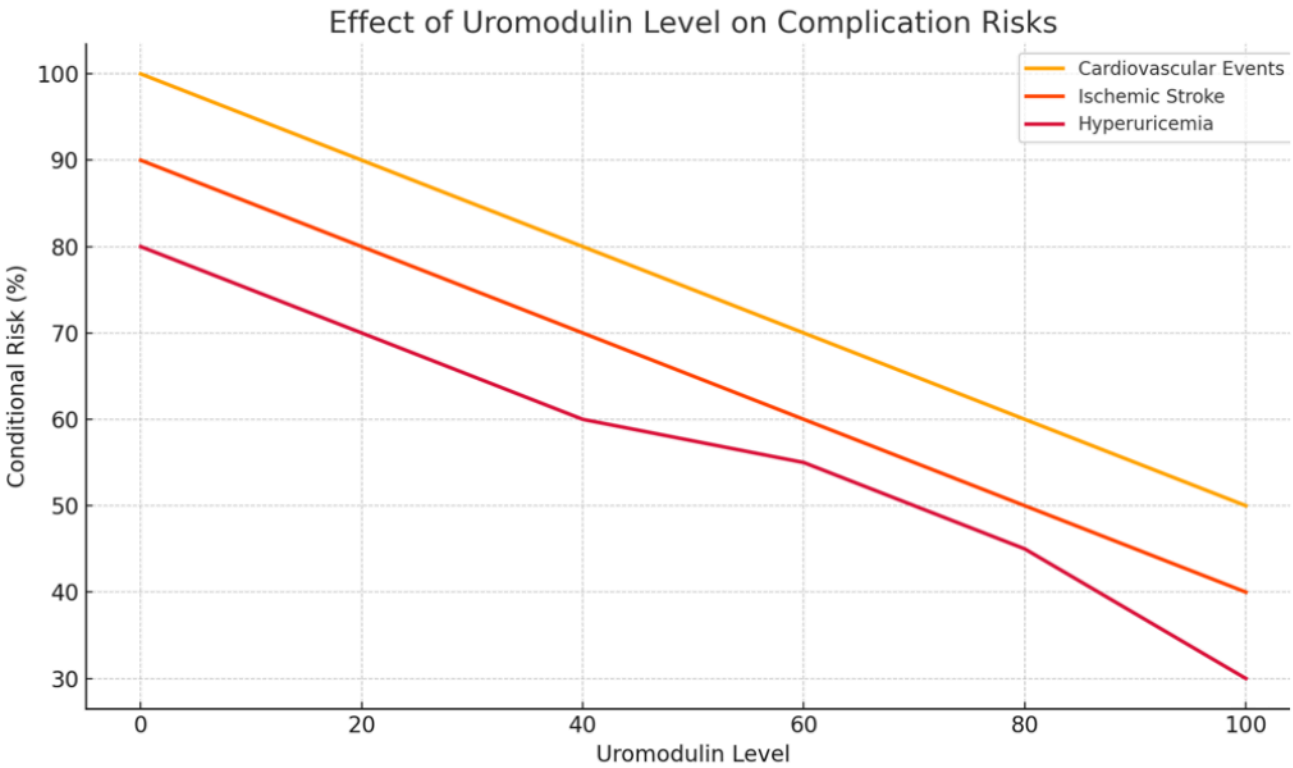

Fig. 1. Inverse relationship between uromodulin level and the risk of cardiovascular events, ischemic stroke, and hyperuricemia [3, 4, 9, 18, 19, 23, 27, 30, 35, 43]

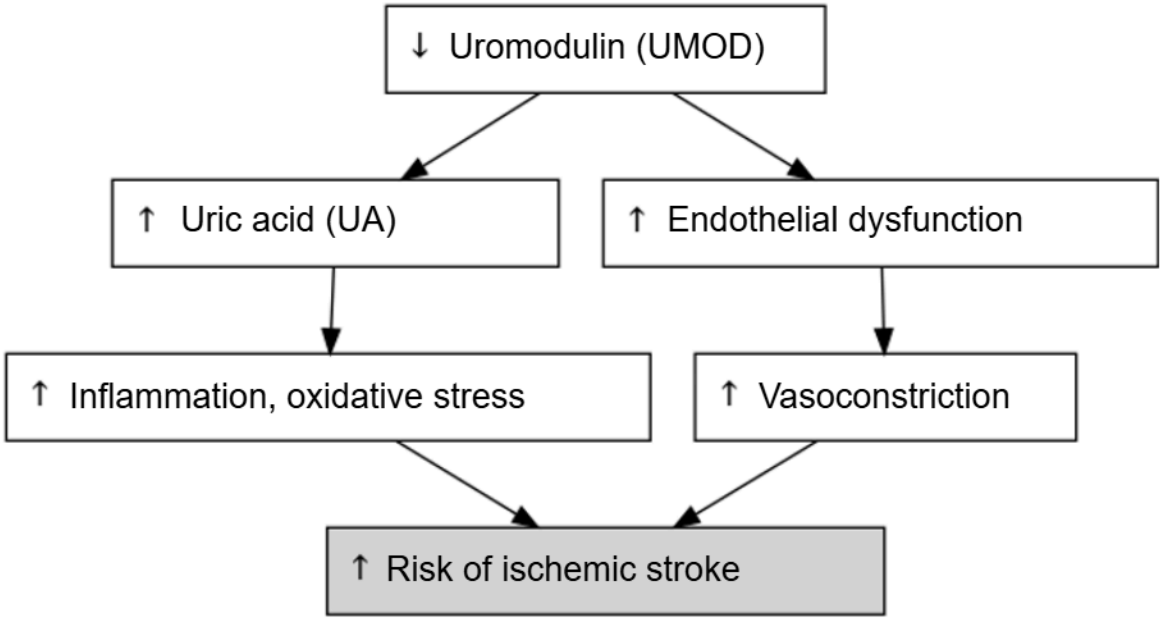

Figure 2. Pathogenetic links between decreased uromodulin, impaired uric acid excretion, and the development of ischemic stroke [3, 14–16, 18–20, 27, 35, 43]

Genetics, metabolomics, and systemic integration

Polymorphisms of the *UMOD* gene play a key role in regulating its expression and are associated with important clinical phenotypes such as arterial hypertension, decreased glomerular filtration rate, hyperuricemia, and stroke risk [2, 5, 16, 97, 8, 19, 22]. In particular, the variants rs4293393 and rs12917707 are associated with elevated serum uric acid levels and decreased urinary excretion [19, 20]. Genetic studies involving more than 300,000 individuals demonstrate a consistent association between the presence of certain *UMOD* allelic variants and an increased risk of developing chronic kidney disease [16].

In addition, metabolomic studies provide insights into the systemic impact of UMOD on various metabolic pathways. Associations have been identified with creatinine, lactate, uric acid levels, as well as biomarkers of inflammation and oxidative stress [49, 50]. This indicates that UMOD functions not only as a marker, but also as an active modulating factor in the pathogenesis of systemic disorders. Such data reinforce the rationale for considering UMOD as a potential target in future therapeutic strategies of personalized medicine.

Systematic reviews, guidelines, and clinical perspective

Over recent years, a number of systematic reviews have been published that thoroughly analyze the role of uromodulin in the pathogenesis of both chronic kidney disease and cardiovascular and cerebrovascular complications. For instance, meta-analyses have shown that low serum UMOD levels are associated with increased cardiovascular mortality and progression of kidney failure independently of other factors [25, 51]. Other systematic studies emphasize the importance of UMOD as a marker of nephron functional reserve, particularly in the preclinical stages of CKD [52].

Recent clinical guidelines discuss the potential of UMOD not only for risk stratification but also for monitoring the effectiveness of nephroprotective therapy. Special attention is given to its application in patients with combined comorbid conditions—arterial hypertension, diabetes mellitus, particularly considering the pathogenetic link between UMOD and diabetes [53], hyperuricemia, and gout [11–13].

Several review articles highlight the potential of UMOD as a marker of cognitive impairment following stroke or as a tool for neurological risk stratification [48, 54, 55]. This approach opens new horizons in studying the link between renal dysfunction and central nervous system disorders. Thus, the accumulated body of evidence supports the consideration of uromodulin not only as an organ-specific indicator, but also as a systemic prognostic marker with potential for clinical implementation.

CONCLUSIONS

UMOD is not only a marker of renal tubular function but also an important systemic bioactive molecule involved in the regulation of vascular homeostasis, uric acid metabolism, endothelial function, and immune responses. Its role extends far beyond nephrology: accumulating evidence indicates associations between UMOD and the risk of hypertension, ischemic stroke, hyperuricemia, cardiovascular events, and cognitive impairment. Genetic and metabolomic studies confirm the link between *UMOD* gene variations and both metabolic and vascular phenotypes, providing further evidence of its systemic impact.

The presence of UMOD in serum and its prognostic value in a range of clinical conditions justify its consideration not only as a diagnostic marker but also as a potential therapeutic target. Further investigation of the molecular mechanisms of UMOD action and its interaction with elements of metabolic and neurovascular regulation opens new horizons for a personalized approach to the prevention and treatment of chronic diseases.

REFERENCES

1. Mary, S., Boder, P., Rossitto, G., et al. (2021). Salt loading decreases urinary excretion and increases intracellular accumulation of uromodulin in stroke-prone spontaneously hypertensive rats. *Clinical science (London)*, 135(24), 2749–2761. https://doi.org/10.1042/cs20211017
2. Mary, S., Small, H. Y., Siwy, J., et al. (2017). Polymerization-incompetent uromodulin in the pregnant stroke-prone spontaneously hypertensive rat. *Hypertension*, 69(5), 910–918. https://doi.org/10.1161/HYPERTENSIONAHA.116.08826
3. Then, C., Then, H. L., Lechner, A., et al. (2020). Serum uromodulin and risk for cardiovascular morbidity and mortality in the community-based KORA F4 study. *Atherosclerosis*, 297, 1–7. https://doi.org/10.1016/j.atherosclerosis.2020.01.030
4. Li, Y., Cheng, Y., Consolato, F., et al. (2022). Genome-wide studies reveal factors associated with circulating uromodulin and its relationships to complex diseases. *JCI Insight,* 7(10), e157035. https://doi.org/10.1172/jci.insight.157035
5. Armstrong, G. K., Frank, J. A., McLouth, C. J., et al. (2021). Commentary: Use of BACTRAC proteomic database-uromodulin protein expression during ischemic stroke. *Journal of Experimental Neurology*, 2(1), 29–33. https://pmc.ncbi.nlm.nih.gov/articles/PMC7990290/
6. Steubl, D., Schneider, M. P., Meiselbach, H., et al. (2020). Association of serum uromodulin with death, cardiovascular events, and kidney failure in CKD. *Clinical Journal of the American Society of Nephrology*, 15(5), 616–624. https://doi.org/10.2215/cjn.11780919
7. Sjaarda, J., Gerstein, H. C., Yusuf, S., et al. (2018). Blood HER2 and uromodulin as causal mediators of CKD. *Journal of the American Society of Nephrology*, 29(4), 1326–1335. https://doi.org/10.1681/asn.2017070812
8. Then, C., Thorand, B., Then, H. L., et al. (2020). Serum uromodulin is inversely associated with arterial hypertension and the vasoconstrictive prohormone CT-proET-1 in the population-based KORA F4 study. *PLoS One*, 15(8), e0237364. https://doi.org/10.1371/journal.pone.0237364

9. Steubl, D., Buzkova, P., Garimella, P. S., et al. (2020). Association of serum uromodulin with mortality and cardiovascular disease in the elderly-the Cardiovascular Health Study. Nephrol Dial Transplant, 35(8), 1399–1405. https://doi.org/10.1093/ndt/gfz008
10. Vasquez-Rios, G., Katz, R., Levitan, E. B., et al. (2023). Urinary biomarkers of kidney tubule health and mortality in persons with ckd and diabetes mellitus. *Kidney360*, 4(9), e1257–e1264. https://doi.org/10.34067/kid.0000000000000226
11. Mary, S., Conti-Ramsden, F., Boder, P., et al. (2024). Pregnancy-associated changes in urinary uromodulin excretion in chronic hypertension. *Journal of Nephrology*, 37(3), 597–610. https://doi.org/10.1007/s40620-023-01830-6
12. Maglinger, B., Frank, J. A., McLouth, C. J., et al. (2021). Proteomic changes in intracranial blood during human ischemic stroke. *Journal of Neurointerventional Surgery*, 13(4), 395–399. https://doi.org/10.1136/neurintsurg-2020-016118
13. Jian, Z., Yuan, C., & Ma, Y. (2022). Blood pressure mediated the effects of urinary uromodulin levels on myocardial infarction: a Mendelian randomization study. *Hypertension*, 79(11), 2430–2438. https://doi.org/10.1161/HYPERTENSIONAHA.122.19670
14. Matafora, V., Zagato, L., Ferrandi, M., et al. (2014). Quantitative proteomics reveals novel therapeutic and diagnostic markers in hypertension. *BBA Clinical*, (2), 79–87. https://doi.org/10.1016/j.bbacli.2014.10.001
15. Amatruda, J. G., Katz, R., Sarnak, M. J., et al. (2022). Biomarkers of kidney tubule disease and risk of end-stage kidney disease in persons with diabetes and CKD. *Kidney International Reports*, 7(7), 1514–1523. https://doi.org/10.1016/j.ekir.2022.03.033
16. Garimella, P. S., Lee, A. K., Ambrosius, W. T., et al. (2019). Markers of kidney tubule function and risk of cardiovascular disease events and mortality in the SPRINT trial. Eur Heart J., 40(42), 3486–3493. https://doi.org/10.1093/eurheartj/ehz392
17. Yap, I. K., Brown, I. J., Chan, Q., et al. (2010). Metabolome-wide association study identifies multiple biomarkers that discriminate north and south Chinese populations at differing risks of cardiovascular disease: INTERMAP study. *Journal of Proteome Research*, 9(12), 6647–54. https://doi.org/10.1021/pr100798r
18. Zhong, J., Cai, H., Zhang, Z., et al. (2024). Serum uric acid and prognosis of ischemic stroke: Cohort study, meta-analysis and Mendelian randomization study. *European Stroke Journal*, 9(1), 235–243. https://doi.org/10.1177/23969873231209620
19. Aliena-Valero, A., Baixauli-Martín, J., Castelló-Ruiz, M., et al. (2021). Effect of uric acid in animal models of ischemic stroke: A systematic review and meta-analysis. *Journal of Cerebral Blood Flow and Metabolism*, 41(4), 707–722. https://doi.org/10.1177/0271678x20967459
20. Zhang, M., Wang, Y., Wang, K., et al. (2021). Association between uric acid and the prognosis of acute ischemic stroke: A systematic review and meta-analysis. *Nutrition, Metabolism, and Cardiovascular Diseases*, 31(11), 3016–3023. https://doi.org/10.1016/j.numecd.2021.07.031
21. Katsiki, N., Dimitriadis, G. D., & Mikhailidis, D. P. (2021). Serum uric acid and diabetes: from pathophysiology to cardiovascular disease. *Current Pharmaceutical Design*, 27(16), 1941–1951. https://doi.org/10.2174/1381612827666210104124320
22. Wang, Z., Lin, Y., Liu, Y., et al. (2016). Serum uric acid levels and outcomes after acute ischemic stroke. *Molecular Neurobiology*, 53(3), 1753–1759. https://doi.org/10.1007/s12035-015-9134-1
23. Zhu, B., Huang, X., Zhang, J., et al. (2025). New perspective on the prediction and treatment of stroke: the role of uric acid. Neurosci Bull, 41(3), 486–500. https://doi.org/10.1007/s12264-024-01301-3
24. Li, M., Wang, H., & Gao, Y. (2023). Serum uric acid levels and recurrence rate of ischemic stroke: a meta-analysis. *Hormone and Metabolic Research*, 55(7), 493–497. https://doi.org/10.1055/a-2091-1951
25. Ae, R., Kanbay, M., & Kuwabara, M. (2020). The causality between the serum uric acid level and stroke. *Hypertension Research*, 43(4), 354–356. https://doi.org/10.1038/s41440-019-0346-z

26. Lei, Z., Cai, J., Hong, H., & Wang, Y. (2019). Serum uric acid level and outcome of patients with ischemic stroke: a systematic review and meta-analysis. Neurologist, 24(4), 121–131. https://doi.org/10.1097/nrl.0000000000000234
27. Zhang, W., Cheng, Z., Fu, F., & Zhan, Z. (2023). Serum uric acid and prognosis in acute ischemic stroke: a dose-response meta-analysis of cohort studies. *Frontiers in Aging Neuroscience*, (15), 1223015. https://doi.org/10.3389/fnagi.2023.1223015
28. Amaro, S., Jiménez-Altayó, F., & Chamorro, Á. (2019). Uric acid therapy for vasculoprotection in acute ischemic stroke. *Brain Circulation*, 5(2), 55–61. https://doi.org/10.4103/bc.bc_1_19
29. Tariq, M. A., Shamim, S. A., Rana, K. F., et al. (2019). Serum uric acid - risk factor for acute ischemic stroke and poor outcomes. *Cureus*, 11(10), e6007. https://doi.org/10.7759/cureus.6007
30. Dong, Y., Shi, H., Chen, X., et al. (2021). Serum uric acid and risk of stroke: a dose-response meta-analysis. *Journal of Clinical Biochemistry and Nutrition*, 68(3), 221–227. https://doi.org/10.3164/jcbn.20-94
31. Liu, Y., Liu, X., Jia, J., et al. (2022). Uric acid and clinical outcomes in young patients with ischemic stroke. *Neuropsychiatric Disease and Treatment*, (18), 2219–2228. https://doi.org/10.2147/ndt.s373493
32. Ren, W., Wu, J., Wu, Z., et al. (2022). Serum uric acid levels at admission could predict the chronic post-stroke fatigue. *Frontiers in Nutrition*, (9), 850355. https://doi.org/10.3389/fnut.2022.850355
33. Qiao, T., Wu, H., & Peng, W. (2021). The relationship between elevated serum uric acid and risk of stroke in adult: an updated and dose-response meta-analysis. *Frontiers in Neurology*, 12, 674398. https://doi.org/10.3389/fneur.2021.674398
34. Yang, S., Liu, H., Fang, X. M., et al. (2024). Signaling pathways in uric acid homeostasis and gout: From pathogenesis to therapeutic interventions. *International Immunopharmacology*, (132), 111932. https://doi.org/10.1016/j.intimp.2024.111932
35. Kim, K. Y., Shin, K. Y., & Chang, K. A. (2022). Potential biomarkers for post-stroke cognitive impairment: a systematic review and meta-analysis. *International Journal of Molecular Sciences*, 23(2), 602. https://doi.org/10.3390/ijms23020602
36. Gao, B., Bao, Y., Meng, M., et al. (2023). Association of serum uric acid with risk of stroke in US adults: A cross-sectional study from NHANES 1999-2020. *Journal of Stroke and Cerebrovascular Diseases*, 32(8), 107206. https://doi.org/10.1016/j.jstrokecerebrovasdis.2023.107206
37. Hu, F., Yang, J., Wu, S., & Yang, C. (2024). Causal relationship between uric acid and stroke: a two-sample mendelian randomization study. *Medicine (Baltimore)*, 103(42), e39591. https://doi.org/10.1097/md.0000000000039591
38. Zhong, C., Zhong, X., Xu, T., et al. (2017). Sex-specific relationship between serum uric acid and risk of stroke: a dose-response meta-analysis of prospective studies. *Journal of the American Heart Association*, 6(4), e005042. https://doi.org/10.1161/jaha.116.005042
39. Tang, H., Mo, J., Chen, Z., et al. (2019). Uric acid contributes to obesity-paradox of the outcome of ischemic stroke. *Frontiers in Neurology*, (10), 1279. https://doi.org/10.3389/fneur.2019.01279
40. Jiménez-Xarrié, E., Pérez, B., Dantas, A. P., et al. (2020). Uric acid treatment after stroke prevents long-term middle cerebral artery remodelling and attenuates brain damage in spontaneously hypertensive rats. *Translational Stroke Research*, 11(6), 1332–1347. https://doi.org/10.1007/s12975-018-0661-8
41. Troyanov, S., Delmas-Frenette, C., Bollée, G., et al. (2016). Clinical, genetic, and urinary factors associated with uromodulin excretion. *Clinical journal of the American Society of Nephrology*, 11(1), 62–69. https://doi.org/10.2215/cjn.04770415
42. Kim, G. H., & Jun, J. B. (2022). Altered serum uric acid levels in kidney disorders. *Life (Basel)*, 12(11), 1891. https://doi.org/10.3390/life12111891
43. Han, J., Liu, Y., Rao, F., et al. (2013). Common genetic variants of the human uromodulin gene regulate transcription and predict plasma uric acid levels. *Kidney International*, 83(4), 733–740. https://doi.org/10.1038/ki.2012.449

44. Wright, A. F., Rudan, I., Hastie, N. D., & Campbell, H. (2010). A 'complexity' of urate transporters. *Kidney International*, 78(5), 446–452. https://doi.org/10.1038/ki.2010.206
45. Yan, B., Zhang, Z. Z., Huang, L. Y., et al. (2012). OIT3 deficiency impairs uric acid reabsorption in renal tubule. *FEBS Letters*, 586(6), 760–765. https://doi.org/10.1016/j.febslet.2012.01.038
46. Nqebelele, N. U., Dickens, C., Dix-Peek, T., et al. (2019). Urinary uromodulin levels and *UMOD* variants in black South Africans with hypertension-attributed chronic kidney disease. *International Journal of Nephrology*, 2019, 8094049. https://doi.org/10.1155/2019/8094049
47. Liu, Y., Goldfarb, D. S., El-Achkar, T. M., et al. (2018). Tamm-Horsfall protein/uromodulin deficiency elicits tubular compensatory responses leading to hypertension and hyperuricemia. *American Journal of Physiology. Renal Physiology*, 314(6), F1062–F1076. https://doi.org/10.1152/ajprenal.00233.2017
48. Wu, C. H., Yang, C. C., Chang, H. W., et al. (2019). Urinary uromodulin/creatinine ratio as a potential clinical biomarker for chronic kidney disease patients with gout: a pilot study. *Medical Principles and Practice*, 28(3), 273–279. https://doi.org/10.1159/000496844
49. Samanta, A., Rahman, S. M., Vasudevan, A., & Banerjee, S. (2023). A novel combination of OHVIRA syndrome and likely causal variant in UMOD gene. *CEN Case Reports*, 12(2), 249–253. https://doi.org/10.1007/s13730-022-00754-7
50. Iguchi, A., Eino, A., Yamazaki, H., et al. (2013). A novel mutation in the uromodulin gene in a Japanese family with a mild phenotype of familial juvenile hyperuricemic nephropathy. *CEN Case Reports*, 2(2), 228–233. https://doi.org/10.1007/s13730-013-0069-y
51. De Luca Canto, G., Pachêco-Pereira, C., Aydinoz, S., et al. (2015). Diagnostic capability of biological markers in assessment of obstructive sleep apnea: a systematic review and meta-analysis. *Journal of Clinical Sleep Medicine*, 11(1), 27–36. https://doi.org/10.5664/jcsm.4358
52. You, R., Zheng, H., Xu, L., et al. (2021). Decreased urinary uromodulin is potentially associated with acute kidney injury: a systematic review and meta-analysis. *Journal of Intensive Care*, 9(1), 70. https://doi.org/10.1186/s40560-021-00584-2
53. Li, X., Meng, X., Timofeeva, M., et al. (2017). Serum uric acid levels and multiple health outcomes: umbrella review of evidence from observational studies, randomised controlled trials, and Mendelian randomisation studies. *BMJ*, 357, j2376. https://doi.org/10.1136/bmj.j2376
54. Liu, Z. Q., Huang, Z. W., Kang, S. L., et al. (2021). Serum uric acid and cardiovascular or all-cause mortality in peritoneal dialysis patients: a systematic review and meta-analysis. *Frontiers in Cardiovascular Medicine*, 8, 751182. https://doi.org/10.3389/fcvm.2021.751182
55. Kang, T., Hu, Y., Huang, X., et al. (2022). Serum uric acid level and all-cause and cardiovascular mortality in peritoneal dialysis patients: A systematic review and dose-response meta-analysis of cohort studies. *PLoS One*, 17(2), e0264340. https://doi.org/10.1371/journal.pone.0264340
56. Ying, H., Yuan, H., Tang, X., et al. (2021). Impact of serum uric acid lowering and contemporary uric acid-lowering therapies on cardiovascular outcomes: a systematic review and meta-analysis. *Frontiers in Cardiovascular Medicine*, (8), 641062. https://doi.org/10.3389/fcvm.2021.641062

Отримано [Received] 22.02.2025

Системна дія уромодуліну: позаниркові ефекти та клінічне значення

Л. Д. Денова, *аспірант,*
ORCID ID: 0000-0002-5678-5885,
Національний університет охорони здоров'я імені П.Л. Шупика, м. Київ, Україна,
Кафедра нефрології і нирковозамісної терапії,
marbua18@gmail.com

С. В. Кушніренко, *доктор медичних наук, професор,*
ORCID ID: 0000-0001-5518-7210,
Національний університет охорони здоров'я імені П.Л. Шупика, м. Київ, Україна,
Кафедра нефрології і нирковозамісної терапії,
stella-alex@i.ua

О. О. Руденко, *невролог,*
ORCID ID: 0009-0006-8794-5438,
Броварська багатопрофільна клінічна лікарня, Бровари, Україна,
rudenkoolga443@gmail.com

Анотація. *Уромодулін (UMOD) — найбільш абундантний білок сечі людини, який традиційно розглядався як маркер функціонального стану ниркових канальців. Проте сучасні наукові дослідження все частіше вказують на його багатогранну системну дію, яка виходить далеко за межі нефрона. Цей білок бере участь у складних механізмах регуляції судинного тонусу, метаболізму сечової кислоти, імунної відповіді, а також функціонування ендотелію. Встановлено його зв'язок із розвитком серцево-судинної патології, ішемічного інсульту, гіперурикемії та метаболічного синдрому. У даній оглядовій роботі проаналізовано широкий спектр експериментальних і клінічних джерел, включаючи метааналізи, систематичні огляди, генетичні та метаболомні дослідження, які підтверджують значну роль UMOD у патогенезі позаниркових захворювань.* ***Висновки.*** *Узагальнені дані дозволяють розглядати UMOD як перспективний біомаркер і потенційний інструмент персоналізованої медицини.*

Ключові слова: *уромодулін; сечова кислота; ішемічний інсульт; гіпертензія; позаниркова дія; біомаркер.*

Автори заявляють про відсутність конфлікту інтересів. [The authors declare no conflict of interest.]

Funding

This publication is a part of the postgraduate research project conducted at the Department of Nephrology and NSET of the P.L. Shupyk National University of Health Care of Ukraine, under the theme: *"Uromodulin excretion and its clinical-laboratory assessment, significance in early diagnosis, renoprotection, and optimization of CKD treatment under molecular stress conditions"*, within the framework of the department's state-funded research project *"Organization and provision of nephrological care under conditions of limited resources and martial law"* (State Registration No. 0123U101260, 2023–2024).

It also includes findings from an initiative-based exploratory research project titled *"Study of the impact of hypouricemic therapy in patients with CKD and justification of optimal treatment strategies"* (State Registration No. 0119U101718, 2019–2025), funded by the state budget of Ukraine.

Цитування (ДСТУ):
Денова Л. Д., Кушніренко С. В., Руденко О. О. Системна дія уромодуліну: позаниркові ефекти та клінічне значення. *Лікарська справа*. 2025. № 1. С. 41–49. https://doi.org/10.31640/LS-2025-1-05

Citation (APA):
Denova, L. D., Kushnirenko, S. V., & Rudenko, O. O. (2025). Systemic action of uromodulin: extra-renal effects and clinical significance. *Likars'ka Sprava*, (1), 41–49. https://doi.org/10.31640/LS-2025-1-05